EXPERIMENT : 7

AIM:

Random Sampling and Sampling Distribution

OBJECTIVE:

To explore random sampling from a population and understand the concept of sampling distribution using Python in Jupyter Notebook.

STEPS:

1. Generate a Population:
    - Create a population of data with a specified distribution (e.g., normal distribution).
2. Random Sampling:
    - Perform random sampling from the population to create multiple samples of different sizes.
    - Compute sample statistics (mean, standard deviation, etc.) for each sample.
3. Sampling Distribution:
    - Plot histograms or density plots of sample statistics (e.g., sample means).
    - Compare the sampling distribution of the sample statistic (mean) with the population distribution.
4. Central Limit Theorem (Optional):
    - Demonstrate the Central Limit Theorem by showing that as sample size increases, the sampling distribution of the sample mean approaches a normal distribution regardless of the population distribution.

CODE:

```
import numpy as np
import matplotlib.pyplot as plt

# Step 1: Generate a population
population_mean = 50
population_std = 10
population_size = 100000

population = np.random.normal(
    population_mean,
    population_std,
    population_size
)

# Step 2: Random sampling
sample_sizes = [20, 80, 100]
num_samples = 500

sample_means = {}

for size in sample_sizes:
    sample_means[size] = []

    for _ in range(num_samples):
        sample = np.random.choice(
            population,
            size=size,
            replace=False
```

```
        )
        sample_means[size].append(np.mean(sample))

# Step 3: Plotting sampling distributions
plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):
    plt.subplot(len(sample_sizes), 1, i + 1)

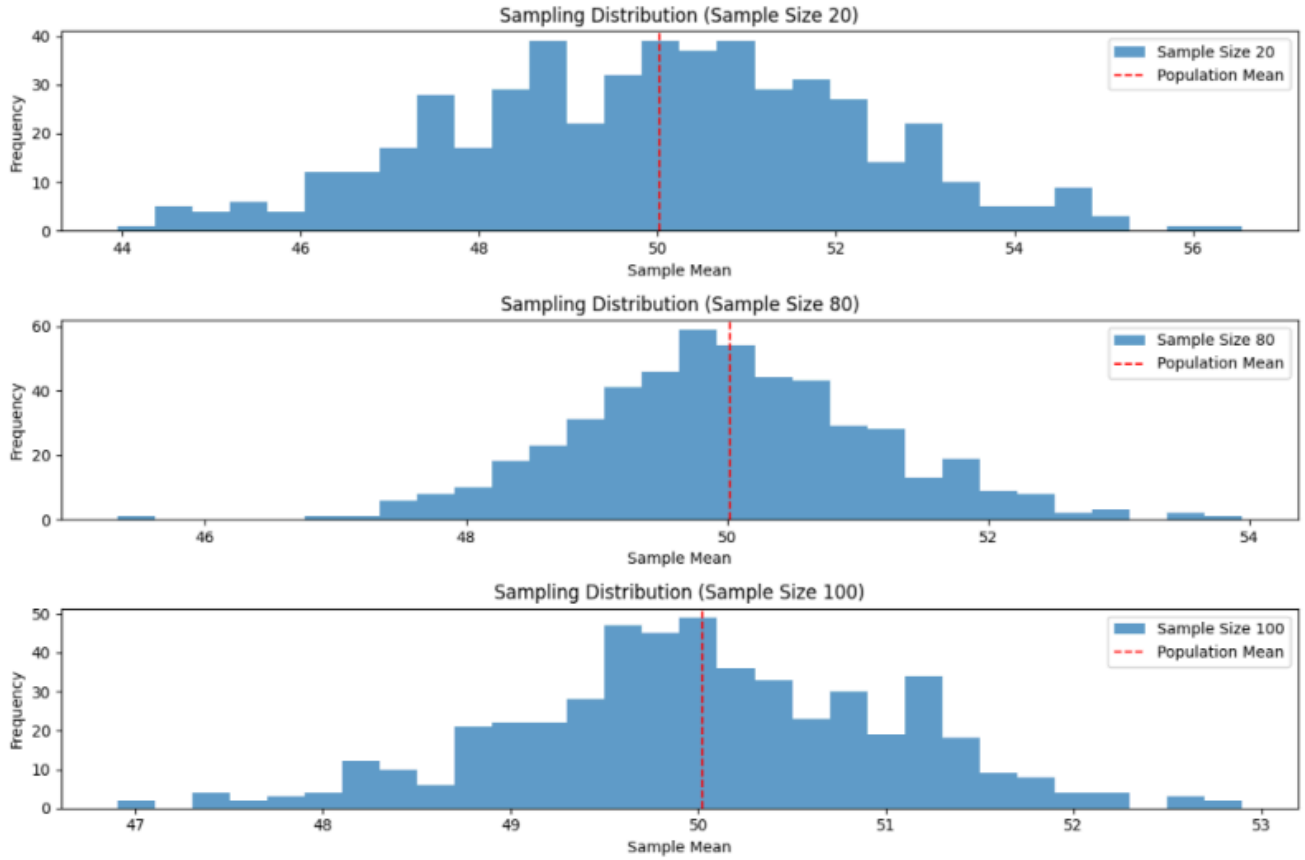
    plt.hist(
        sample_means[size],
        bins=30,
        alpha=0.7,
        label=f'Sample Size {size}'
    )

    plt.axvline(
        np.mean(population),
        color='red',
        linestyle='dashed',
        linewidth=1.3,
        label='Population Mean'
    )

    plt.title(f'Sampling Distribution (Sample Size {size})')
    plt.xlabel('Sample Mean')
    plt.ylabel('Frequency')
    plt.legend()

plt.tight_layout()
plt.show()
```

OUTPUT:



RESULT:

Random sampling was successfully performed from the given population using Python in Jupyter Notebook. The sampling distribution was generated and analyzed to understand how sample statistics vary across different samples.